MARA × Soluna

Buy the project, not the company

An independent M&A case study · Data as of March 28, 2024

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The answer

Rebuilt from the FY2023 10-K, the whole-company deal does not pay. The value is in one project.

–$20.3M	NPV of a $4.00/share tender for all of Soluna (TEV $35.9M vs $17.6M value to MARA)
$6.0M	Maximum all-cash price for the 50 MW Dorothy 2 project: $4.0M at close + $2.0M earn-out
+$3.6M	NPV to MARA on a $9.5M value (range $5.6–15.0M); 0% dilution; no HSR filing

Gate: fixed-price build at or below $0.50M/MW. Breakeven is $0.55M/MW.

Sources: company 10-Ks; Yahoo Finance prices 3/28/2024; author model (model/MARA_Soluna_model.xlsx).

MARA Holdings (NASDAQ: MARA)

Large, liquid, and renting all of its capacity. In Q1-2024 it started buying operating sites at about $0.45M per MW.

FY2023	Value
Revenue	$387.5M
Net income	$261.2M
Adjusted EBITDA (incl. $331.5M bitcoin fair-value gains)	$419.9M
Total margin excluding D&A	$164.2M
Bitcoin mined / held	12,852 / 15,126 BTC
Energized / avg. operational hashrate	24.7 / 19.4 EH/s
Cash / convertible notes (principal)	$357.3M / $330.7M
Self-hosted capacity at year-end	None (~193,000 hosted miners)
Generate Capital sites, Jan-2024	390 MW for $179.0M
Garden City, TX, Mar-2024	200 MW for $87.3M
Market cap / EV, 3/28/2024	$5.48B / $5.45B (14.1× revenue)

Sources: MARA FY2023 10-K and 10-K/A; Applied Digital 8-K (4/1/2024); Yahoo Finance. EV uses 12/31/2023 shares.

Soluna Holdings (NASDAQ: SLNH)

75 MW operating, but only 41.5 MW of that is Soluna's. Plus a 50 MW shovel-ready expansion, a going-concern opinion and an $85M preferred claim.

SITES AT 12/31/2023

Site	MW	Soluna share	Activity	FY23 rev / cash cost
Sophie, KY	25	100%	Hosting	$6.0M / $3.2M
Dorothy 1A, TX	25	15%	Hosting	$6.9M / $4.4M
Dorothy 1B, TX	25	51%	Mining	$6.8M / $3.4M
Dorothy 2, TX	50	100%	Shovel-ready	—
Kati, TX	166	100%	Development	—

FY2023 FINANCIALS

Item	Value	Item	Value
Revenue	$21.1M	Cash (unrestricted)	$6.4M
Operating loss	$(20.2)M	Debt principal	$19.6M
Net loss (consolidated)	$(27.7)M	Series A claim (pref. + arrears)	$85.1M
Adj. EBITDA (Q4: +$1.0M)	$(3.5)M	Common shares (3/28/24)	2.84M

Revenue: crypto $10.6M, hosting $10.2M, demand response $0.3M. Attributable FY2023 revenue: $10.5M. Convertible notes mature July 25, 2024; the NYDIG loan is in default.

Source: Soluna Holdings FY2023 Form 10-K (filed 4/1/2024). Yahoo Finance.

Strategic rationale

Owning capacity makes sense for MARA. Owning Soluna's capital structure does not.

WHAT HOLDS

- Self-hosting keeps the host's margin: about $18/MWh at Dorothy
- Curtailment pays: ERCOT demand response worth about $32k per MW-yr
- Dorothy 2 is shovel-ready: power and land in place, could energize in 2025

WHAT DOESN'T

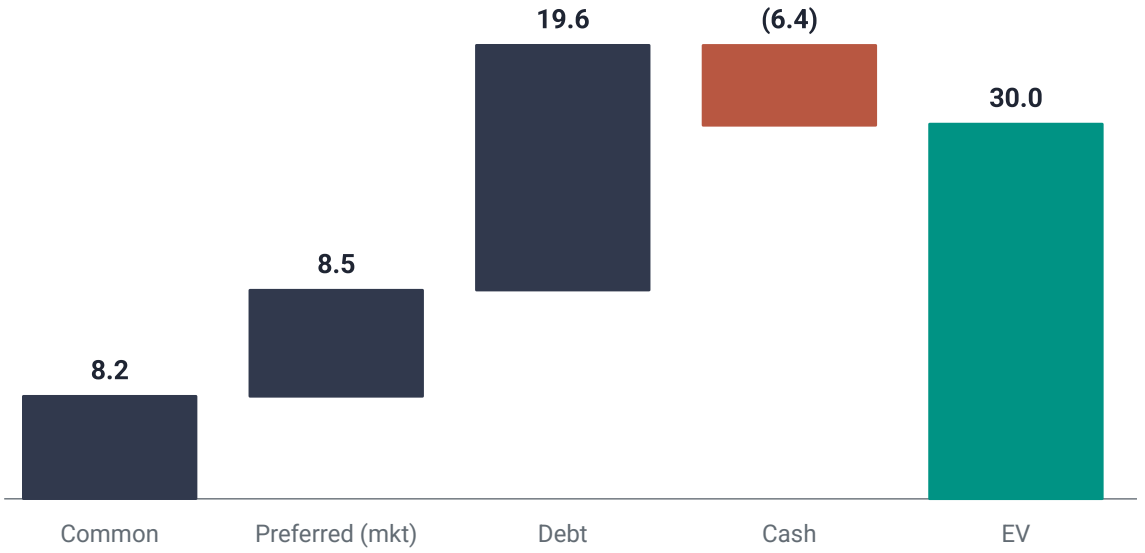
- MARA would not own Dorothy outright: Soluna holds 15% of 1A and 51% of 1B
- Moving MARA miners into full sites displaces paying customers, so the saving is zero
- The $85M preferred claim and $19.6M of 2024 debt use up any control value
- ESG benefit is real but not quantified

Hosting margin estimated from the FY2023 10-K site table (D1A, annualized, 10% haircut). Demand response at the Dec-2023 run-rate with a 50% haircut (assumption).

Capital structure and EV

Market value attributable to Soluna is $30.0M, or $0.72M per attributable MW. The preferred is priced at 10 cents on its claim.

ATTRIBUTABLE EV, $M



Claim	Market	Face / claim
Series A preferred	$8.5M	$85.1M
Consolidated EV (incl. NCI)	$56.5M	2.7× revenue

Prices: Yahoo Finance close 3/28/2024 (SLNH $2.88; SLNHP $2.79). Balance sheet: Source: Soluna Holdings FY2023 Form 10-K (filed 4/1/2024).

Trading comparables

Peers trade at a median 7.9× revenue. Applied to Soluna that gives $48–84M, which is far above its own $30M market value. Shown for context, not relied on.

Company	Mkt cap	EV	EV / Rev	EH/s	EV / EH/s
MARA	$5,483M	$5,451M	14.1×	24.7	$221M
RIOT	$2,825M	$2,229M	7.9×	12.4	$180M
CLSK	$3,936M	$3,902M	18.2×	10.0	$390M
CORZ	$645M	$1,131M	2.3×	16.9	$67M
HUT	$982M	$1,141M	n.m.	6.3	$181M
IREN	$456M	$365M	n.m.	5.6	$65M
BITF	$745M	$661M	4.5×	6.5	$102M
WULF	$728M	$808M	11.7×	5.5	$147M
Peer median (ex-MARA)			**7.9×**		**$147M**
25th – 75th percentile			4.5–11.7×		$84–180M

EV/EBITDA is not used because FY2023 EBITDA includes bitcoin fair-value gains (ASU 2023-08). HUT and IREN revenue is n.m. (transition period / June fiscal year).

Sources: FY2023 10-K/10-KT/40-F/6-K filings; CORZ 3/31/2024 10-Q (post-emergence); CLSK LTM Dec-23. Prices Yahoo 3/28/2024. EV = mkt cap + debt + pref − cash.

Precedent transactions

MARA's own Q1-2024 site purchases give the best benchmark: $0.44–0.46M per MW. On that basis Soluna is worth $22.5–25.9M.

Date	Buyer / asset	MW	Price	$M / MW
Jan-2024	MARA / Generate Capital sites (TX, NE)	390	$179.0M	0.46
Mar-2024	MARA / Applied Digital Garden City, TX	200	$87.3M	0.44

IMPLIED VALUE OF SOLUNA, $M

Component	Low	High
41.5 attributable operating MW × $0.44–0.46M	18.1	19.0
Dorothy 2: 50 MW × 20–30% of operating $/MW	4.4	6.9
Precedent-implied value	**22.5**	**25.9**
Market EV, 3/28/2024	30.0	30.0

This implies $0.09–0.14M per MW for shovel-ready Dorothy 2 rights, which anchors the price recommended later ($0.12M/MW).

Sources: MARA FY2023 10-K/A (Generate Capital); Applied Digital 8-K filed 4/1/2024 (Garden City; up to $34M price reduction if the expansion is not approved). Shovel-ready discount is an assumption.

Discount rate

A 20% WACC. This is a micro-cap with minority-owned sites and a going-concern opinion, and it carries close to MARA's own beta.

Input	Value	Basis
Risk-free rate	4.20%	10-year Treasury, 3/28/2024
Equity risk premium	4.60%	Damodaran implied, Jan-2024
Unlevered beta	2.42	Median of 7 peers
Target debt / capital	25%	Assumption
Relevered beta	3.23	No tax shield (NOLs)
Company-specific premium	3.0%	Assumption
Cost of equity	22.1%	
Pre-tax cost of debt	14.0%	NYDIG loan rate (proxy)
WACC	**20.0%**	**20.04% before rounding**
Cross-check: MARA cost of equity	20.7%	β 3.58

Sources: FRED DGS10 (3/28/2024); Damodaran implied ERP (Jan-2024); betas from 2-year weekly Yahoo returns vs S&P 500, unlevered at 21% tax. Assumptions labelled.

DCF: Soluna under MARA

With G&A cut and Dorothy 2 built, the attributable business is worth $17.6M. Standalone, overhead consumes the site cash flow.

$M	2024*	2025	2026	2027	2028
Existing sites (attrib.)	5.1	5.4	5.6	5.6	5.6
Dorothy 2 (100%)	0.0	2.1	8.4	8.4	8.4
Corporate G&A	(5.6)	(2.6)	(2.7)	(2.7)	(2.8)
EBITDA	**(0.6)**	**4.9**	**11.3**	**11.2**	**11.1**
Taxes, one-time & maint.	(2.8)	(0.9)	(2.1)	(2.1)	(2.1)
D2 build capex	(11.3)	(11.3)	0.0	0.0	0.0
Unlevered FCF	**(14.6)**	**(7.3)**	**9.2**	**9.2**	**9.1**

ATTRIBUTABLE EV, $M (WACC 20%, g 2%)

Case	Base	WACC 18–22%, g 1–3%
Standalone (existing sites)	(31.1)	(36.2) to (27.3)
Under MARA (whole company)	**17.6**	**12.4 to 25.0**
Dorothy 2 only, before price	9.5	5.6 to 15.0

Terminal value at 4.6× 2028 EBITDA. Under-MARA EV at bitcoin $50k / $100k: $14.8M / $21.3M.

Bitcoin flat at $70,745; hashprice $55.0 (Q2–Q4 2024) and $44.4/PH/day (2025); D2 build $0.45M/MW; 21% tax under MARA. 2024 = Q2–Q4. Mid-period discounting. All assumptions in model.

Sensitivity

Value moves more with the discount rate and build cost than with the bitcoin price. Build cost is the one variable the deal terms can control.

UNDER-MARA EV ($M): WACC × g

WACC	g = 1%	g = 2%	g = 3%
18%	21.2	23.0	25.0
19%	18.6	20.1	21.8
20%	16.3	17.6	19.1
21%	14.2	15.4	16.7
22%	12.4	13.4	14.5

BITCOIN PRICE

BTC	Under MARA EV
$50k	14.8
$60k	16.1
$70.7k (3/28/24)	17.6
$85k	19.5
$100k	21.3

DOROTHY 2 DEAL NPV ($M): BUILD COST × HOSTING MARGIN

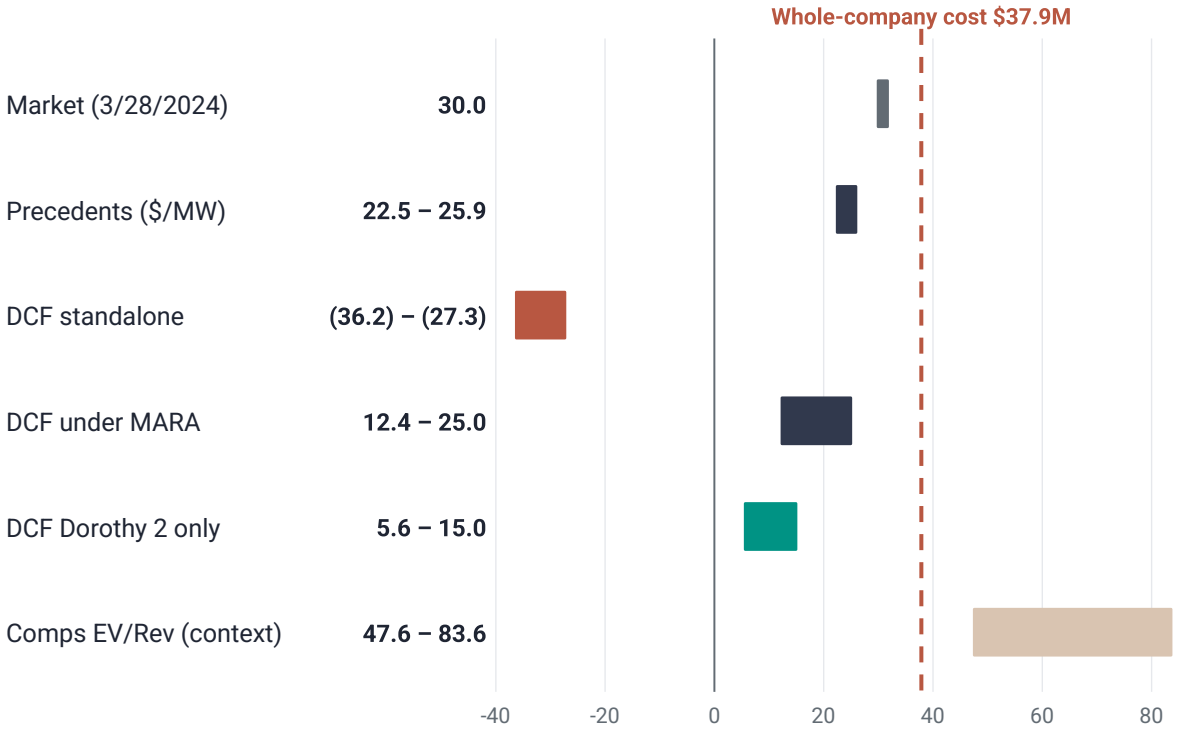
Build $M/MW	$0.100M/MW-yr	$0.136M (base)	$0.170M/MW-yr
$0.35	1.2	7.3	13.1
$0.45	(2.5)	3.6	9.4
$0.55	(6.2)	(0.1)	5.7
$0.65	(10.0)	(3.8)	2.0

Breakeven: $0.55M/MW build cost or $0.115M/MW-yr margin (16% below base).

Author model (model/MARA_Soluna_model.xlsx). WACC × g grid is live in DCF_Control; bitcoin and build-cost grids are exported from model.py to the Deal sheet.

Valuation summary

Every method except comps values Soluna below the $37.9M cost of the whole-company deal.



Dorothy 2 price: up to $6.0M (PV incl. fees $5.9M) against $5.6–15.0M of value.

Attributable EV, $M. Comps shown, not relied on. DCF ranges: WACC 18–22%, g 1–3%. Market: 3/28/2024.

Tested and rejected: buy all of Soluna

A $4.00 tender costs $37.9M against $17.6M of value. Even at $0 for the common, NPV is –$8.3M.

Use of funds	Terms	$M
Common equity	$4.00 × 2.91M FD shares; +38.9% spot, +18.7% VWAP	11.6
Series A preferred	$3.50 tender, +25.4% to market	10.7
Convertible notes	Greater of par or as-converted	9.0
NYDIG + Navitas loans	Repaid at principal	10.9
Less: cash acquired		(6.4)
Transaction EV	**1.7× FY2023 revenue**	**35.9**
Fees	Assumption	2.0
Total cost		**37.9**
DCF value to MARA		17.6
NPV		**(20.3)**

Stress test: residual G&A $1.5–3.5M × WACC 18–22% gives NPV of –$9.8M to –$28.6M. Paying in stock (0.99M MARA shares, 0.37% dilution) changes the currency, not the value.

Spot $2.88; 30-day VWAP $3.37 (Yahoo). Preferred change-of-control terms per Series A 424B5 (conversion capped at 0.2817 common shares per preferred, post-split). NYDIG settlement value not verified.

Recommended: buy Dorothy 2 for cash

A $6.0M ceiling with $2.0M paid only on energization. MARA funds the $22.5M build from cash on hand.

Term	Proposal
Asset	Dorothy 2, 50 MW, Silverton TX: site, power and interconnection rights, permits, engineering
Consideration	All cash: $4.0M at close + $2.0M earn-out if ≥45 MW energized by 9/30/2025
Price / MW	$0.12M, inside the $0.09–0.14M precedent-implied range
Value / NPV	$9.5M value; PV of price + fees $5.9M; NPV +$3.6M
Financing	Cash on hand: $29.0M total incl. build, 8% of YE-2023 cash; no new debt
Ownership / EPS	0% dilution; –0.3% (2025E), +0.8% (2026E) of FY2023 EPS
Conditions	Power rights assigned (ERCOT/utility consents); fixed-price EPC ≤ $0.50M/MW; lien releases; Soluna board approval
Other	O&M contract with Soluna on market terms; ROFO on Kati; no HSR filing

Why cash: Soluna needs liquidity ($6.4M of cash against notes due July 2024), so cash at close is the most valuable thing MARA can offer. The earn-out moves timing risk to the seller. Stock for a $6M asset adds cost, not value.

HSR: FTC 2024 size-of-transaction threshold $119.5M. All terms are the author's proposal; price, earn-out and EPC gate are assumptions tested in the model.

Synergies, quantified

The real synergy is $8.4M a year of self-hosting margin on new capacity. Moving miners into Soluna's full sites is worth zero.

Synergy	Whole company	D2 asset deal	Basis
Corporate G&A removed	$7.7M/yr	—	$10.3M less $2.6M residual; $2.5M one-time cost
Self-hosting margin, new MW	$8.4M/yr	$8.4M/yr	($0.136M + $0.032M DR) × 50 MW, from 2026
MARA miners into existing sites	$0	$0	Displaces paying hosting customers
AI / HPC conversion	Not valued	Not valued	No contracted demand; upside only

Why the G&A saving does not rescue the whole-company deal: it is already inside the $17.6M under-MARA value. Clearing the preferred, notes and loans (net of cash, plus fees) costs about $26M before anything goes to the common.

Margins from the FY2023 10-K site table; G&A figures are assumptions (standalone $10.0M in 2024 +3%/yr; residual $2.5M). The original deck's $8–12M cost + $5–10M revenue synergies counted hosting margin twice.

Key risks

Build cost is the main risk, so it is a closing condition.

Risk	Effect	Mitigation
Build cost overrun	–$3.7M NPV per +$0.10M/MW; breakeven $0.55M	Fixed-price EPC ≤ $0.50M/MW as a condition
Hosting margin after halving	Breakeven $0.115M/MW-yr (–16%)	Self-host; demand response income
Power rights not assignable	Deal fails	Consents are conditions; no cash before close
Seller distress (going concern, NYDIG default)	Lien or fraudulent-transfer challenge	Fair price vs precedents; lien releases; earn-out holdback
SLC / Navitas rights over D2	Block or delay	Diligence item (unverified)
ERCOT curtailment / policy	Lower uptime, demand response income	85% uptime; DR at 50% haircut
Opportunity cost	Operating MW at $0.44–0.46M	Keep it a bolt-on; all-in $0.57M/MW

Sensitivities from the author model. Partner rights over Dorothy 2 are not disclosed in the FY2023 10-K and are flagged as a diligence item.

Recommendation

1. Do not buy Soluna Holdings.

2. Buy Dorothy 2 for up to $6.0M cash, behind a $0.50M/MW build gate.

3. Keep buying operating sites; Dorothy 2 adds to that program and does not replace it.

WHAT HAPPENED NEXT (NOT USED IN THE VALUATION)

- Soluna built Dorothy 2 itself. Spring Lane Capital committed up to $30M for 48 MW (~$0.63M/MW, July 2024), and it was completed in November 2025. At that cost the asset deal would have had an NPV of about –$2.9M, so the EPC gate was the right protection.
- MARA was already a Soluna hosting customer at Project Sophie (agreement dated January 3, 2024, disclosed March 2025). Capacity could be had by contract.
- SLNH shares outstanding grew from 2.8M to 244.6M by August 2026. The stock closed at $0.72 on October 14, 2026 (–75%).
- Soluna revenue was $38.0M in FY2024 and $29.7M in FY2025, with net losses of $(63.3)M and $(53.4)M.

Hindsight sources: Soluna FY2024 and FY2025 10-Ks; Soluna 10-Q (Q2-2026); Soluna 8-K Ex.10.1 (Mar-2025); MARA FY2024/FY2025 10-Ks; Yahoo Finance 10/14/2026.

Sources and method

Every figure is traced in figures.json (value, unit, as-of date, source or "assumption"). The model has live formulas.

- SEC EDGAR FY2023 filings: Soluna 10-K; MARA 10-K and 10-K/A; Riot 10-K; Core Scientific 10-K and Q1-24 10-Q; Hut 8 10-KT; TeraWulf 10-K; Bitfarms 40-F; CleanSpark 10-K and 10-Q; IREN 6-K
- Soluna Series A preferred 424B5 (change-of-control terms); Applied Digital 8-K (Garden City sale)
- Market data: Yahoo Finance daily prices and volumes (3/28/2024); FRED DGS10 and DTB3; Damodaran implied ERP (Jan-2024); blockchain.com difficulty and fees
- FTC 2024 HSR thresholds
- Model: MARA_Soluna_model.xlsx (Inputs, Comps, WACC, Ops, three DCFs, Deal, Football)
- Independent case study by Chafic Charafeddine. No non-public information was used, and no company or adviser reviewed it.